Giap Nguyen Trong

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Research Interests

My research focuses on the dynamics, estimation, and control of underactuated and flexible robotic systems. I am interested in reduced-order models, such as pseudo-rigid-body (PRB) chains, that stay accurate under large deformation yet are light enough to run inside a real-time control loop, and in pairing them with nonlinear and sampling-based model predictive control and with state estimation (UKF and observers). A recurring theme is control under minimal sensing: the deformation of a flexible body is often not measured, so it must be estimated, and model mismatch can bias both the estimate and the controller acting on it. I also study learning-based predictors of dynamic behavior from sparse observations, with the aim of exploiting a flexible body's passive dynamics for tasks such as swing-up, tool manipulation, and locomotion. My work spans simulation and hardware, from a flexible inverted pendulum rig to humanoid platforms.

Education

B.Sc. in Mechatronics Engineering Sep 2022 – May 2027 (expected)
Hanoi University of Science and Technology (HUST), Hanoi, Vietnam
Cumulative GPA: 2.76 / 4.0; semester GPA 2.88, 3.00, 2.88 over the last three terms.
Core coursework: Analytical Mechanics, Modern Control Theory, Nonlinear Systems, Optimal Control.
English: IELTS Academic scheduled for January 2027 (expected overall band 6.0–6.5).

Research & Professional Experience

- **AiRA Laboratory (HUST) & RMIT Collaborative Robotics, Hanoi**
  Undergraduate Robotics Researcher Aug 2024 – Present
  - *Minimal-sensing phantom bending* (hardware; journal draft). On a cart-pole with a flexible TPU beam sensed only at the cart and base, identified spurious "phantom" bending estimates in reduced-order observers; designed a dual-channel UKF and a real-time NMPC that balance the beam in simulation and on the physical rig.
  - *Dynamic tool manipulation* (ICRA 2027; simulation and learning). Learned a model that predicts a flexible rod's endpoint response from sparse visual markers, and embedded it in sampling-based MPC to bring the tool tip beyond its quasi-static reach in simulation.
  - *Global swing-up via NMPC* (simulation). Built a two-segment PRB model and a `jaxopt` NMPC that exploits the whip effect to swing up an underactuated flexible inverted pendulum at low solve latency.
  - *High-fidelity PRB modeling* (simulation). Developed a floating-frame PRB with closed-form joint stiffness identification that matches continuum references closely at high resolution.
  - *Vibration locomotion* (simulation). Mapped contact dynamics of a Flix-Walker in MuJoCo and built a posture-alignment navigation strategy that reached every benchmark goal under payload and friction changes.

- **Magic Wave, Hanoi**
  Engineer Apr 2026 – Present
  - *Whole-body control* on a Unitree humanoid for locomotion balance and upper-body teleoperation tracking; trained and validated in simulation, then deployed on the physical robot in real time.

Publications & Manuscripts

P5. **Giap Nguyen Trong**, Son Dinh Bao, Long, Hai-Nguyen Nguyen, and Chien Truong-Quoc. 2027. Beyond Static Reach: Learning to Wield a Flexible Tool from Sparse Visual Feedback. *IEEE International Conference on Robotics and Automation (ICRA 2027)*, under review.

P4. **Giap Nguyen Trong** and Chien Truong-Quoc. In preparation. Spurious Bending Estimates under Minimal Sensing in Flexible Inverted Pendulums. Manuscript drafted; experiments ongoing.

P3. Vuong Duc Minh, **Giap Nguyen Trong**, Minh Tuan Nguyen Thai, and Chien Truong-Quoc. 2026. Motion-Primitive Navigation for a Flix-Walker under Payload and Friction Variations. In *Proceedings of IFToMM ISRM*.

P2. **Giap Nguyen Trong** and Chien Truong-Quoc. 2026. Global Swing-up and Vibration Suppression of a Flexible Inverted Pendulum via NMPC and PRB Modeling. Presented at the *HUST Student Scientific Research Conference*.

P1. **Giap Nguyen Trong**, Chien Truong-Quoc, and Hoang Nguyen Quang. 2025. High-Fidelity Modeling of the Flexible Inverted Pendulum via Pseudo-Rigid-Body (PRB). In *Proceedings of the 8th International Conference on Engineering Mechanics and Automation (ICEMA 2025)*, Hanoi.

Technical Skills

Control and estimation: nonlinear dynamics, minimal PRB / continuum beams, NMPC (SQP-RTI, MPPI), UKF and observers.

Software: Python, C/C++, ROS / ROS 2, MuJoCo, JAX / `jaxopt`, `acados`, CasADi, MATLAB/Simulink, Linux, LaTeX.

Hardware: real-time control, Unitree platforms, sensor interfaces (encoders, potentiometers), 3D printing (TPU/PLA).

Additional

Mentoring: guide junior students on robotics and control topics.

Languages: Vietnamese (native); English (IELTS Academic scheduled January 2027, expected 6.0–6.5).